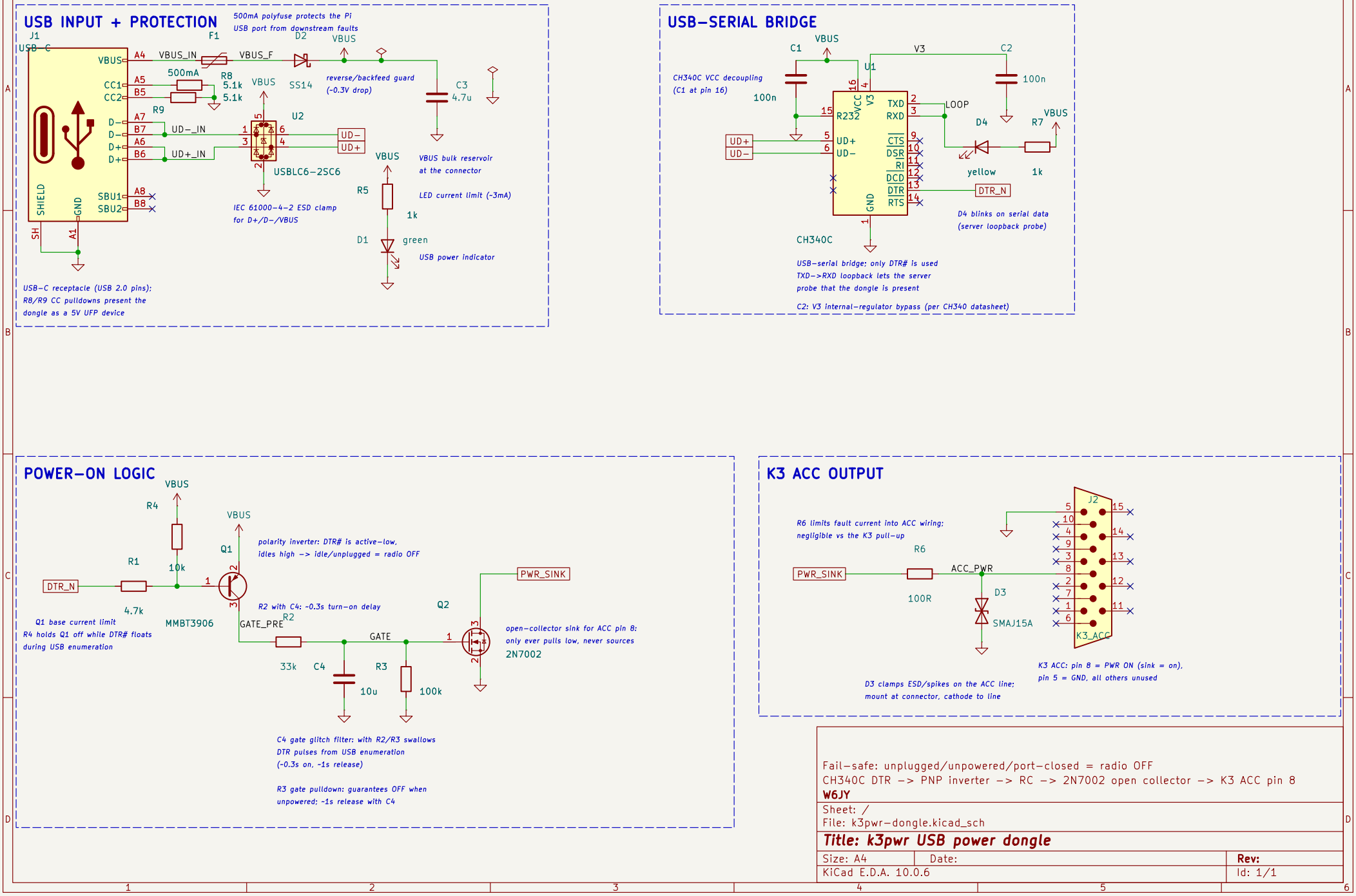

USB INPUT + PROTECTION
500mA polyfuse protects the Pi
USB port from downstream faults
J1
USB_C
F1
D2
VBUS
VBUS_IN
VBUS_F
500mA
R8
5.1k
R9
5.1k
SS14
reverse/backfeed guard
(~0.3V drop)
C3
4.7u
U2
USBLC6-2SC6
UD-_IN
UD+_IN
UD-
UD+
IEC 61000-4-2 ESD clamp
for D+/D-/VBUS
VBUS bulk reservoir
at the connector
R5
1k
LED current limit (~3mA)
D1
green
USB power indicator
SHIELD
GND
SH
USB-C receptacle (USB 2.0 pins):
R8/R9 CC pulldowns present the
dongle as a 5V UFP device
USB-SERIAL BRIDGE
C1
100n
V3
C2
100n
U1
CH340C
CH340C VCC decoupling
(C1 at pin 16)
R232
TXD
RXD
LOOP
D4
R7
1k
yellow
DTR_N
CTS
DSR
RI
DCD
DTR
RTS
D4 blinks on serial data
(server loopback probe)
USB-serial bridge: only DTR# is used
TXD->RXD loopback lets the server
probe that the dongle is present
C2: V3 internal-regulator bypass (per CH340 datasheet)
POWER-ON LOGIC
R4
10k
R1
4.7k
Q1
MMBT3906
polarity inverter: DTR# is active-low,
idles high -> idle/unplugged = radio OFF
Q1 base current limit
R4 holds Q1 off while DTR# floats
during USB enumeration
R2
33k
GATE_PRE
GATE
C4
10u
R3
100k
R2 with C4: ~0.3s turn-on delay
Q2
2N7002
PWR_SINK
open-collector sink for ACC pin 8:
only ever pulls low, never sources
C4 gate glitch filter: with R2/R3 swallows
DTR pulses from USB enumeration
(~0.3s on, ~1s release)
R3 gate pulldown: guarantees OFF when
unpowered; ~1s release with C4
K3 ACC OUTPUT
R6 limits fault current into ACC wiring;
negligible vs the K3 pull-up
R6
100R
ACC_PWR
D3
SMAJ15A
J2
K3_ACC
D3 clamps ESD/spikes on the ACC line;
mount at connector, cathode to line
K3 ACC: pin 8 = PWR ON (sink = on),
pin 5 = GND, all others unused
Fail-safe: unplugged/unpowered/port-closed = radio OFF
CH340C DTR -> PNP inverter -> RC -> 2N7002 open collector -> K3 ACC pin 8
W6JY
Sheet: /
File: k3pwr-dongle.kicad_sch
Title: k3pwr USB power dongle
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 1/1